

D 2

$$y = x e^x$$

$$y' = (1)(e^x) + (x)(e^x) = e^x + x e^x = e^x (1+x)$$

$$y'' = (e^x)(1+x) + (e^x)(1) = e^x + x e^x + e^x = 2e^x + x e^x = e^x (2+x)$$

$$e^x (2+x) \geq 0 \quad \left\{ \begin{array}{l} e^x \geq 0 \\ 2+x \geq 0 \end{array} \right\} \quad \forall x$$

$$x \text{ compreso per } x \in [-2, +\infty)$$

D 4

$$y = \sqrt{\frac{x}{x^2+1}}$$

$$y' = \frac{1}{2 \sqrt{\frac{x}{x^2+1}}} \cdot \frac{(x^2+1) - (x)(2x)}{(x^2+1)^2} = \frac{1}{2 \sqrt{\frac{x}{x^2+1}}} \cdot \frac{x^2+1-2x^2}{(x^2+1)^2} = \frac{1}{2 \sqrt{\frac{x}{x^2+1}}} \cdot \frac{-x^2+1}{(x^2+1)^2} = \frac{\sqrt{x^2+1}}{2 \sqrt{x}} \cdot \frac{-x^2+1}{(x^2+1)^2} = \frac{\sqrt{x^2+1} (-x^2+1)}{2 \sqrt{x} (x^2+1)^2}$$

$$\sqrt{x^2+1} (-x^2+1) < 0 \Rightarrow -x^2+1 < 0 \Rightarrow x^2-1 > 0 \Rightarrow x^2 > 1 \Rightarrow x < -1 \vee x > 1 \quad \text{per } x < -1 \text{ i non e' definita}$$

D 3

$$a_n = \frac{n^2 4^n - 6^n}{n 3^n - n^n} \sim \frac{-6^n}{-n^n} = \frac{6^n}{n^n} = 0$$

D₃

$$e^{x-2} < 1 \Rightarrow e^{x-2} < e^0 \Rightarrow x^2 - 2 < 0 \Rightarrow x^2 < 2 \Rightarrow -\sqrt{2} < x < \sqrt{2}$$

D₇

$$\int_1^2 \frac{x}{x+1} = \left[x - \ln(x+1) \right]_1^2 = (2 - \ln(3)) - (1 - \ln(2)) = 2 - \ln(3) + (-1 + \ln(2)) = 1 - \ln(3) + \ln(2) = 1 - (\ln(3) - \ln(2)) = 1 - \ln(3/2)$$

$$\int \frac{x}{x+1} = \int \frac{x+1-1}{x+1} = \int \frac{y-1}{y} = \int \frac{1}{y} - \frac{1}{y} = \int 1 - \frac{1}{y} = x - \ln(y) = x - \ln(x+1)$$

D₁

$$y = x+1 \\ dy = dx \\ y = \begin{cases} \frac{\ln(1+2x)}{x} & -1/2 < x < 0 \\ k(x-1)^2 - \frac{1}{x+1} & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} \frac{\ln(1+2x)}{x} = \frac{\ln(1+2x)}{2x} \cdot \frac{2x}{x} = 2$$

$$\lim_{x \rightarrow 0^+} k(0-1)^2 - \frac{1}{0+1} = k(-1)^2 - 1 = k - 1$$

$$2 = k - 1$$

$$3 = k$$

continua per $k=3$

D₆

$$\sum \frac{(n+1)!}{10n^2} \quad \text{he somme positive}$$

Eg 3

$$\int \frac{x}{e^x} = \int x e^{-x} = (x)(-e^{-x}) - \int -e^{-x} = x(-e^{-x}) + \int e^{-x} = -x e^{-x} - e^{-x} = -e^{-x}(x+1) + C$$

$$F(0) = -e^{-0}(0+1) = -1(0+1) = -1$$

$$-1 + C = -1 \Rightarrow C = 0$$

$$-e^{-x}(x+1)$$

2)

Dato un successione a termini positivi

$$x \sqrt[n]{a_n} = l \in \mathbb{R}$$

$$\text{se } l > 1 \text{ on diverge a } +\infty$$

$$\text{se } l < 1 \text{ on converge a } 0$$

$$\text{se } l = 1 \text{ il criterio è inconcludente}$$

3)

$$\lim_{n \rightarrow +\infty} e^{-x}(x+1) = \frac{x+1}{e^x} = 0 \quad \text{però che}$$

infiniti

Ex 2

$$y = \frac{e^x}{x^2 - 3}$$

DOMINIO

$$x^2 - 3 > 0 \Rightarrow x^2 > 3 \Rightarrow x < -\sqrt{3} \vee x > \sqrt{3}$$

$$x \in \mathbb{R} \setminus \{\pm\sqrt{3}\}$$

SIMMETRIA

$f(x)$ non ha simmetria

Limiti

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - 3} = +\infty \quad \begin{array}{l} \text{parabola} \\ \text{inf} \end{array} \quad \text{as. verticale}$$

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - 3} \cdot \frac{1}{x} = +\infty + \infty \quad \text{no as. obliqua}$$

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2 - 3} = \frac{0}{-\infty} = 0 \quad \text{as. orizzontale}$$

$$\lim_{x \rightarrow (-\sqrt{3})^-} \frac{e^{-\sqrt{3}}}{3 - 3} = \frac{e^{-\sqrt{3}}}{0} = +\infty \quad \text{as. verticale}$$

$$\lim_{x \rightarrow (\sqrt{3})^+} \frac{e^{\sqrt{3}}}{3 - 3} = \frac{e^{\sqrt{3}}}{0} = +\infty \quad \text{as. verticale}$$

SEGNO

$$\frac{e^x}{x^2-3} \geq 0 \quad \left\{ \begin{array}{l} e^x \geq 0 \\ x^2-3 \geq 0 \end{array} \right. \quad \left\{ \begin{array}{l} \forall x \\ x^2 \geq 3 \end{array} \right. \quad x < -\sqrt{3} \vee x > \sqrt{3}$$

$$f(x) \geq 0 \text{ per } x \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, +\infty)$$

MONOTONIA

$$y = \frac{e^x}{x^2-3}$$

$$y' = \frac{e^x(x^2-3) - (e^x)(2x)}{(x^2-3)^2} = \frac{x^2 e^x - 3e^x - 2x e^x}{(x^2-3)^2} = \frac{e^x(x^2-2x-3)}{(x^2-3)^2}$$

$$\frac{e^x(x^2-2x-3)}{(x^2-3)^2} > 0 \quad \left\{ \begin{array}{l} e^x > 0 \\ x^2-2x-3 > 0 \end{array} \right. \quad \left\{ \begin{array}{l} \forall x \\ x < -1 \vee x > 3 \end{array} \right.$$

$$x \neq \pm \sqrt{3}$$

$$f(x) \text{ crescente per } x \in (-\infty, -\sqrt{3}) \cup (-\sqrt{3}, -1) \cup (3, +\infty)$$

MASSIMI E MINIMI

$$\frac{e^x(x^2-2x-3)}{(x^2-3)^2} = 0$$

$$e^x(x^2-2x-3) = 0$$

$$x^2-2x-3=0$$

$$x = -1 \vee x = 3$$

$$f(3) \text{ minimo locale}$$

$$f(-1) \text{ massimo locale}$$

